

# Video Conferencing Platform

## ( Zoom Clone)

*SDE Fullstack Assignment*

### Description

Build a functional **video conferencing web application clone of Zoom web app** that replicates Zoom's design, user experience, and core meeting workflows.

The platform should allow users to create meetings, join meetings, schedule meetings, manage participants, and experience a clean, professional interface as original Zoom.

Your implementation should visually and functionally feel as a modern **Zoom Meeting Platform**.

**Note:** Your application should match the original application in terms of UI and UX, look and feel should be exactly the same.

### AI Tools Usage

You are **allowed and encouraged** to use AI tools such as ChatGPT, Claude, GitHub Copilot, Cursor, or any other AI assistants for development. However, you **must understand every line of code** you submit and be prepared to explain your implementation decisions during the evaluation interview.

### Technical Stack

- **Frontend:** Next.js (Single Page Application)
- **Backend:** Python with FastAPI/Django
- **Database:** SQLite (design your own schema)

### Core Features (Must Have)

#### 1. Landing Dashboard

Create a Zoom dashboard/homepage.

##### Requirements:

- Clean professional Zoom UI
  - Navbar with profile/settings placeholders
  - Buttons for:
    - New Meeting
    - Join Meeting
    - Schedule Meeting
  - Upcoming meetings section
  - Recent meetings section
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## 2. Instant Meeting Creation

### Requirements:

- Create a new meeting instantly
- Generate unique Meeting ID
- Generate shareable invite link
- Redirect user to meeting room

## 3. Join Meeting

### Requirements:

- Join using Meeting ID or invite link
- Enter display name before joining
- Validate meeting existence

## 4. Schedule Meetings

### Requirements:

- Create scheduled meetings
- Title / Description
- Date & Time picker
- Duration
- Auto-generate meeting link
- Store in database
- Show in Upcoming Meetings section

## Good to Have (Bonus)

- Responsive design (mobile, tablet, desktop)
- User authentication (Login/Signup)
- Host controls (mute all / remove participant)

## Important Notes

- **UI Design:** Your application should totally resemble Zoom's design. Study Zoom's UI carefully before starting.
- **No Login Required:** Assume a default user is logged in. Focus on the functionality rather than authentication.
- **Sample Data:** Seed your database.
- **Database Design:** Design your own database schema. This will be evaluated.
- **README File:** Include setup instructions, tech stack used, and any assumptions made.
- **Original Work:** Plagiarism from existing repositories will result in immediate disqualification.

## Submission

- Upload your code to GitHub and ensure the repository is **public**
- Deploy your application (Vercel, Netlify, Render, Railway, or any cloud service)
- Submit both the GitHub repository link and the deployed application link

## Evaluation Criteria

| Criteria           | What We Look For                                   |
|--------------------|----------------------------------------------------|
| Functionality      | All core features working correctly                |
| UI/UX              | Visual similarity to Zoom's design and UX patterns |
| Database Design    | Well-structured schema with proper relationships   |
| Code Quality       | Clean, readable, and well-organized code           |
| Code Modularity    | Proper separation of concerns, reusable components |
| Code Understanding | Ability to explain your code during evaluation     |

## Timeline

**Submission Deadline:** 1 days from the date of receiving this assignment

*Good luck!*